MIL-S-63302(AR)
AMENDMENT 1
9 March 1981

MILITARY SPECIFICATION

SAFETY AND ARMING ASSEMBLIES
FOR
MINES: M74 AND M75

This Amendment forms a part of Military Specification MIL-S-63302, dated 24 July 1979

PAGE 1

Change "XM74 and XM75" in the title to "M74 and M75"; and wherever it appears.

PAGE 2

Add to "MIL-L-63246 - Leads, Explosive ....." "PA521".

PAGE 4

3.3, change from ".... of MIL-STD-331 except .... in 4.5.8.1." to "... of MIL-STD-331 except as follows: Side output from two HMX loaded cups placed end to end shall produce .020 inch minimum (min) dent in an aluminum witness plate when tested as specified in 4.5.8.1. Side output from a single HMX loaded cup shall produce a 0.013 inch min dent."

3.5.b, change "... slider and its explosive trains to the in-line ....." to "... slider to move to the in-line .....".

3.6.1, delete, "in the aluminum witness plate,".

Change table as follows:

TEMPERATURE	DEPTH OF DENT
-35° -5°F	.005 inch min
+75° ±10°F	.005 inch min
+145° ±5°F	.005 inch min

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PAGE 5

3.6.2, line 5 delete, ".. in a steel confining ring (witness block) ..".

Last line - change, "Dwg. 93287516" to "Dwg. 9328756".

3.8, change "0.015 ohm." to "0.15 ohm."

PAGE 9

Cable Flexible (Dwg 9329826), delete: Salt Spray and associated entries.

PAGE 13

Change: "S&A Assy (Dwg. 9292756)" to "S&A Assy (Dwg. 9328756)".

Change: "MFD assemblies and shield" to "MDF assemblies and shield".

PAGE 14

Notes: Change "Half the sample shall ..... facing down." to "** Half of the sample units shall be functioned with lock plate facing up and the remaining sample units with lock plate facing down."

Add ** to the MDF Train Function column entitled: No. of Sample Units.

For item: Main charge lead function (4 places), change "4.5.18.2" to "4.5.18.1".

PAGE 16

4.4.2.1, Major 105, change "width" to "with".

Add Major 112 - Length to main lead blast hole, 4 places - 0.40% - 3.2 - Gage.

Minor 202: Change "rang" to "tang".

Delete Minor 204 in its entirety.

PAGE 18

4.4.2.3: In Notes (a), change "Inspection Level S-3" to "Inspection Level S-1".

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PAGE 19

4.4.2.4, Major 103 change, "Thickness of plate" to "Thickness of plate - detail A of drawing".

In Notes (b), change "Inspection Level S-3" to "Inspection Level S-1".

PAGE 20

Delete Major 121 and No. 122 in their entirety.

PAGE 23

4.4.2.7: Major 105, change to read:

"Chamfer missing, 2 places - 0.40% - 3.2 - Visual."

PAGE 24

4.4.2.8: Notes (a), change "Inspection Level S-3" to "Inspection Level S-1".

PAGE 25

4.4.2.9: Delete Major 101 in its entirety.

PAGE 28

4.4.2.12: Major 112, change to read:

"True position of largest holes."

Major 114, change to read:

"True position of smallest hole with Datum A and B."

PAGE 30

4.4.2.14: Major 119, change to read:

"Length of cutout for lock plate."

Add a second sentence to Notes:

"The features may be gaged before the chromate finish is added - provided that - after parts are chromated, they still meet requirements of drawing."

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PAGE 31

4.4.2.14: Major 126, change to read:

"Wall thickness at end of slider cavity."

PAGE 33

4.4.2.15: Add a second sentence to Notes:

"The features may be gaged before the chromate finish is added - provided that - after parts are chromated, they still meet requirements of drawing."

PAGE 38

4.4.2.18: Major 103, change "imcomplete" to "incomplete".

Major 106, change to read:

"Length of roll crimped cup, min, 2 places."

Major 109, change to read:

"Spacing present between MDF and HMX, 2 places - 100% - 3.2 - 4.5.11."

Add: Major 112 - Inclusions, voids or impurities - 100% - 3.2 - 4.5.11."

Add this sentence to Notes (a):

"The samples from Major 107 may be used in output test of Major 110."

ADD Note (b): "Results from #110 shall be recorded and forwarded to DRDAR-QAA-M (See 6.5)."

PAGE 40

4.4.2.20: Delete Major 103 in its entirety.

PAGE 42

4.4.2.22: Major 113, change to read:

"Height between tabs."

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PAGE 44

4.4.2.23: Major 101, change to read:

"Explosive lead, PA517, inverted, 2 places."

PAGE 45

4.4.2.24: Major 102, change to read:

"Explosive lead, PA517, inverted, 2 places."

In Notes (a), change: "Inspection Level I." to "Inspection Level S-3."

Delete last sentence of Notes (a): "Any failure ..... to this test."

PAGE 47

4.4.2.26: Delete Major 103 in its entirety.

PAGE 53

4.4.2.32: Add: "Major 109 - Length of smallest diameter - 0.40% - 3.2 - Gage".

PAGE 57

4.4.2.37: Notes, delete "*See 4.5.2" and substitute "*See 4.5.3".

PAGE 58

4.4.2.37: Change title to read:

"Safety and Arming Assembly (Prior to assembling Upper and Lower Housing Assemblies)".

Delete Critical 2 in its entirety.

Delete Critical 5 in its entirety.

PAGE 59

4.4.2.38: Change title to read:

"Safety and Arming Assembly (Prior to assembling Upper and Lower Housing Assemblies)".

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Delete Critical 2 in its entirety.

Delete Critical 5 in its entirety.

PAGE 60

4.4.2.39: Critical 1, change "3.5.7.2" to "4.5.7.2".

PAGE 61

4.4.2.40: Critical 1, change "3.5.7.2" to "4.5.7.2".

PAGE 62

4.4.2.41: Major 101, change to read:

"Actuator spacer, blast shield assy and flexible cable assy missing or incorrectly installed."

PAGE 63

4.4.2.42: Major 103, change to read:

"Blast shield assy, actuator spacer, MDF spacer and flexible cable assy missing or incorrectly installed."

PAGE 64

4.4.2.43: Critical 3, change to read: "Slider release pin and shear pin missing or improperly assembled".

Also add: Critical 4, centrifugal lock binding, sticking or restricted in travel to armed position and/or return to the safe position, 2 places. (c), 100%, 3.2, Visual/Manual also add Note (c) prior to installing lock plate, each centrifugal lock must be moved to the fully armed position and observation made of free return of lock to safe position.

PAGE 65

4.4.2.44: Change Critical 1 to read: "Eyelet ground open circuit, detonators or micro piston actuator not commonly shorted to S&A body."

Critical 3, change to read: ""Slider release pin and shear pin missing or improperly assembled".

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Also add: Critical 4, centrifugal lock binding, sticking or restricted in travel to armed position and/or return to the safe position, 2 places. (c), 100%, 3.2, Visual/Manual also add Note (c) prior to installing lock plate, each centrifugal lock must be moved to the fully armed position and observation made of free return of lock to safe position.

PAGE 66

4.4.2.45: For item: Final Arming, change "4.5.7.2" to "4.5.17.2".

Add Note 3: "Results from in-line explosive functioning shall be recorded and forwarded to DRDAR-QAA-M (See 6.5)."

PAGE 67

4.4.2.46: For item: Main charge train functioning, change "4.5.18.2" to "4.5.18.1".

Add Note 3: "Results from in-line explosive functioning shall be recorded and forwarded to DRDAR-QAA-M (see 6.5)."

PAGE 73

4.5.8.1, change "Witness plate, 5/8 X 1 1/4 X 1 1/4 inches," to "Witness plates, 5/8 X 2 X 18 inches for series end to end output and 1/8 X 1/2 X 3/4 inches for single cup output,.."

Add to end of paragraph:

"A maximum of ten (10) test samples may be tested end to end at one time on a single 5/8 X 2 X 18 inch witness plate. MDF shall be formed at a $90^{\circ} \pm 5^{\circ}$ angle with a .047 inch inside bend radius."

4.5.8.2, change last sentence to read: "Acceptable assemblies subjected to this test may be used in output test of 4.5.8.1."

4.5.9, change last sentence to read:

"This is a non-destructive test and acceptable parts meeting no-movement requirement of drawing may be returned to the lot."

4.5.11, 4.5.11.1 and 4.5.11.2, combine into single 4.5.11:

"4.5.11 Neutron radiographic examination. The MDF assembly, Dwg 9328723, shall be subjected to neutron radiographic examination. Observation shall be made for compliance with requirements of the drawing. This is a non-destructive test and acceptable parts may be returned to the lot."

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PAGE 74

4.5.13, change last sentence to read:

"This is a non-destructive test and acceptable parts may be returned to the lot."

4.5.15.3, add the following:

"NOTE: The S&A assembly may be tested in standard shipping (service item) package in lieu of rigid mounting in a fixture."

PAGE 75

4.5.17.1: Change:

"The S&A test ..... about its center line," to "The S&A test samples from 4.5.16 shall be placed in an approved fixture and spin machine, and subjected to a spin of 2500 RPM max about its center line, ..."

4.5.17.2: Change:

"The sample S&A assemblies form 4.5.17.1 ...." to "The sample S&A assemblies from 4.5.17.1 ...."

4.5.17.2.a, delete in its entirety and substitute:

"An aluminum plate, 1/2 X 3/4 X 1/8 inch thick, shall be secured at the side face of the MDF cap preselected for functioning. Material for the aluminum witness plate shall be as specified in 4.5.8.1 (Dwg 9328756 only)."

4.5.17.2.b, delete in its entirety and substitute:

"The S&A assembly under test shall be inclosed by four (4) steel test plates located at the output of each of the four (4) main explosive leads. Each test plate shall be approximately 1 1/8 X 1 1/8 X 5/8 inch of cold drawn or cold rolled steel, per ASTM-A-108, C1018 or C1020, Rockwell hardness B80 - B95. As an alternate, a cylinder of the above material may be used. The cylinder with an outside diameter of 2.00 inches and an inside diameter of 1.168 + .005 inches with a height sufficient to contain the output shall be secured to preclude displacement during test."

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PAGE 76

4.5.17.2.f, change "... the energy from a one hundred .... volts max." to "... the energy from an eighty-two (82) ± 5% microfarad capacitor charged to five (5) ± 0.1% volts d.c."

4.5.18.1, delete in its entirety and substitute:

"The S&A assemblies previously armed and fitted with witness plates, see 4.5.17.2.a, b, shall be functioned by subjecting the parallel M100 and PA506 delay detonators to the energy from a 82 ± 5% microfarad capacitor charged to 5.0 ± 0.1% volts d.c. Observation shall be made for compliance with requirements of 3.6.1 and 3.6.2. The 20 ms to 50 ms delay time of the delay detonator PA506 shall be recorded."

4.5.18.2, delete in its entirety and substitute:

"The S&A assemblies previously armed and fitted with steel witness plates, see 4.5.17.2.b, shall be functioned by subjecting the M100 detonator to the energy from an eighty-two (82) ± 5% microfarad capacitor charged to five (5) ± 0.1% volts d.c. Observation shall be made for compliance with 3.6.2.

PAGE 77

6.5, delete in its entirety and substitute:

"Submission of ammunition data cards and performance data. In addition to the normal distribution of records, one copy of ammunition data cards and performance data in-line explosive performance data (4.4.2.45, 4.4.2.46) and MDF assembly output test (4.4.2.18) shall be forwarded to Commander, ARRADCOM, ATTN: DRDAR-QAA-M, Dover, NJ 07801.

Custodian:
Army-AR

Preparing Activity:
Army-AR

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